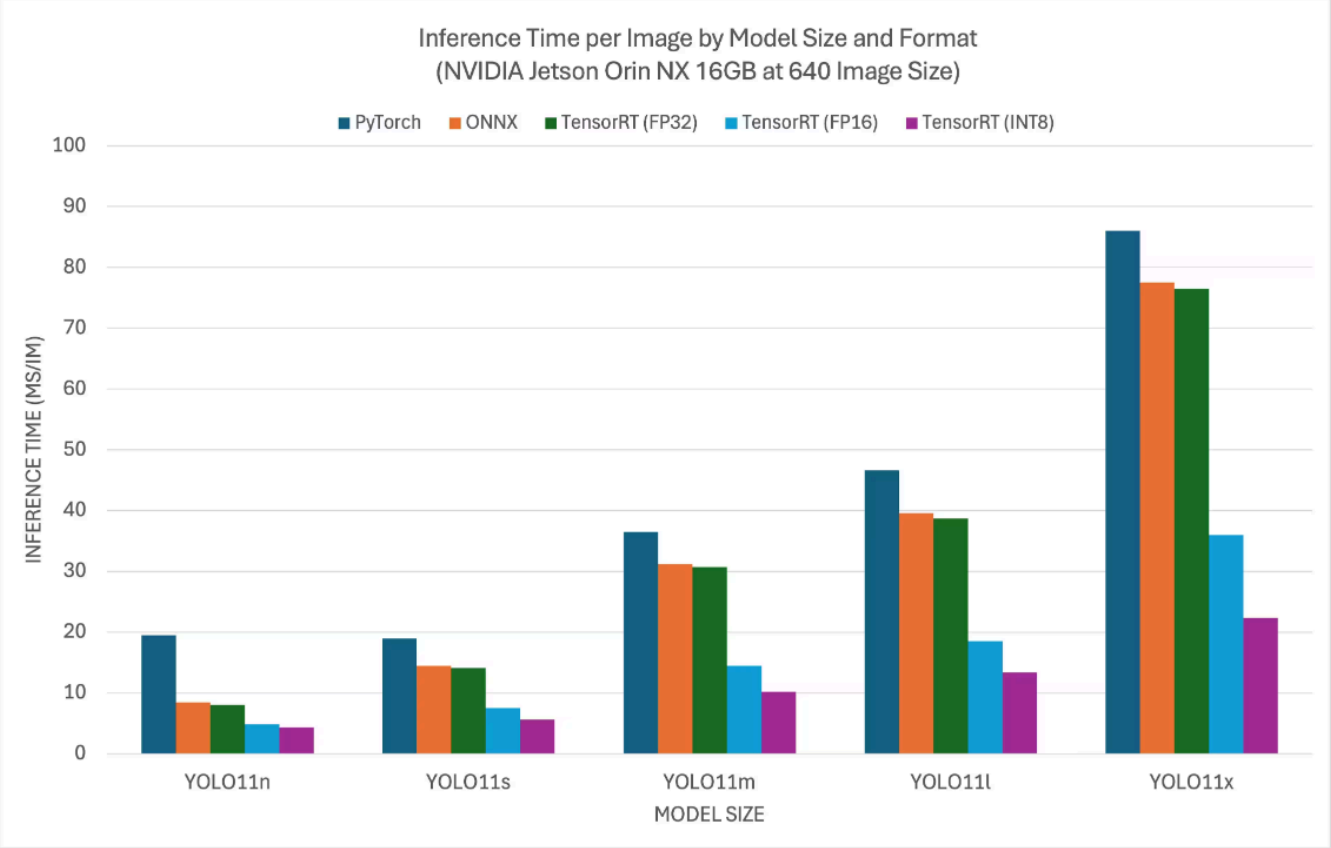
Model conversion

Model conversion

- 1. Jetson AGX Orin YOLO11 (benchmark)
- 2. Enable the maximum performance of the mainboard
  - 2.1. Enable the MAX power mode
  - 2.2. Enable the Jetson clocks
- 3. Model conversion
  - 3.1. CLI: pt → onnx → engine
  - 3.2. Python: pt → onnx → engine
- 4. Model prediction
  - CLI usage
- Frequently Asked Questions
  - ERROR: onnxslim
- References

1. Jetson AGX Orin YOLO11 (benchmark)

The YOLO11 benchmark data comes from the Ultralytics team, which tests models in multiple different formats (data is for reference only)



2. Enable the maximum performance of the mainboard

2.1. Enable the MAX power mode

Enabling the MAX Power Mode (the maximum power mode) on the Jetson ensures that all the CPU and GPU cores are turned on:

```
sudo nvpmodel -m 0
```

2.2. Enable the Jetson clocks

Enabling the Jetson Clocks ensures that all the CPU and GPU cores run at the maximum frequency:

```
sudo jetson_clocks
```

3. Model conversion

According to the test parameters of the models in different formats provided by the Ultralytics team, we can find that the inference performance is the best when using TensorRT!

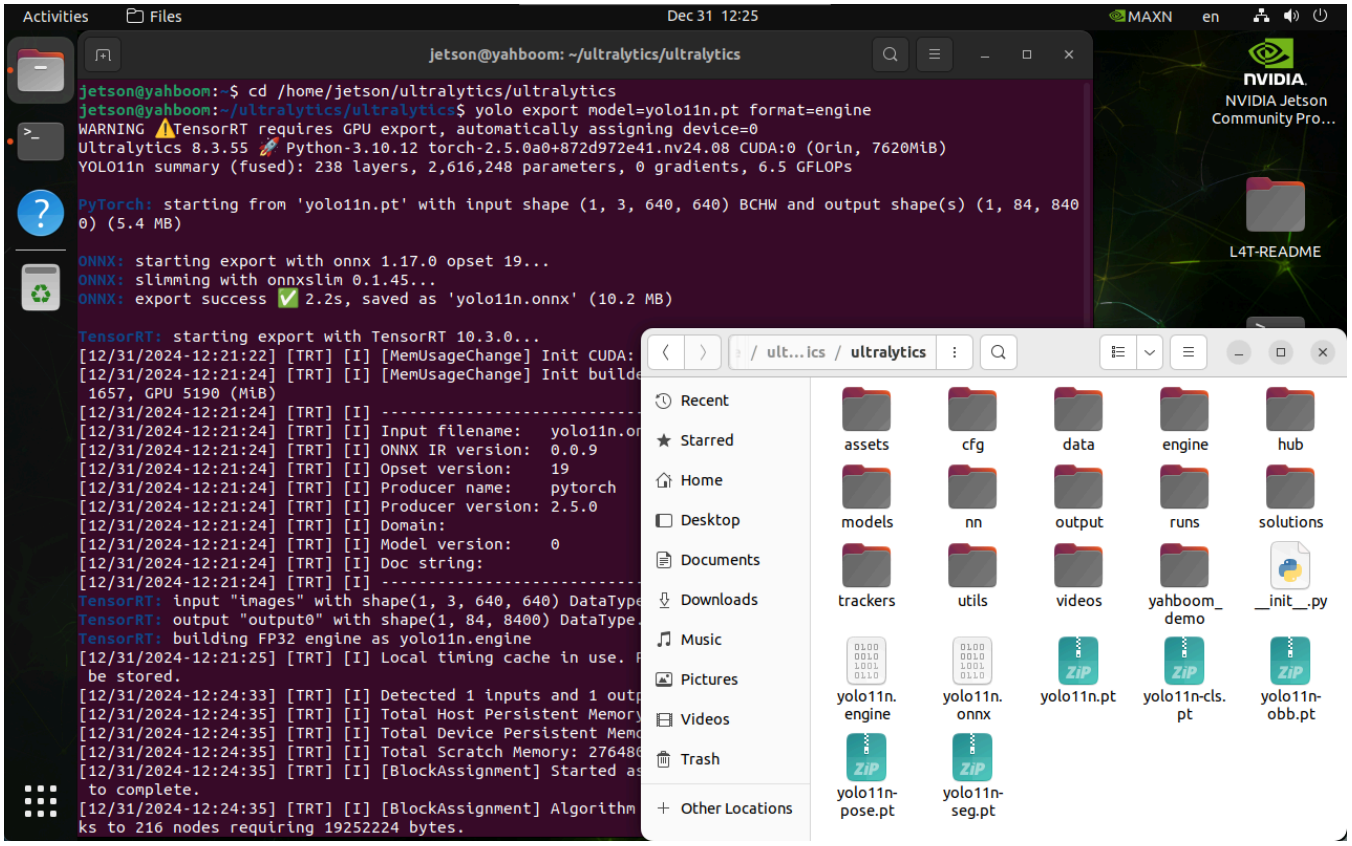
```
The first time you use the export mode of YOLO11, some dependencies will be automatically
installed. Just wait for the automatic installation to complete!
```

3.1. CLI: pt → onnx → engine

Convert the PyTorch format model to TensorRT: the conversion process will automatically generate an ONNX model

```
cd /home/jetson/ultralytics/ultralytics
```

```
yolo export model=yolo11n.pt format=engine
# yolo export model=yolo11n-seg.pt format=engine
# yolo export model=yolo11n-pose.pt format=engine
# yolo export model=yolo11n-cls.pt format=engine
# yolo export model=yolo11n-obb.pt format=engine
```



3.2. Python: pt → onnx → engine

Convert the PyTorch format model to TensorRT: the conversion process will automatically generate an ONNX model

```
cd /home/jetson/ultralytics/ultralytics/yahboom_demo
```

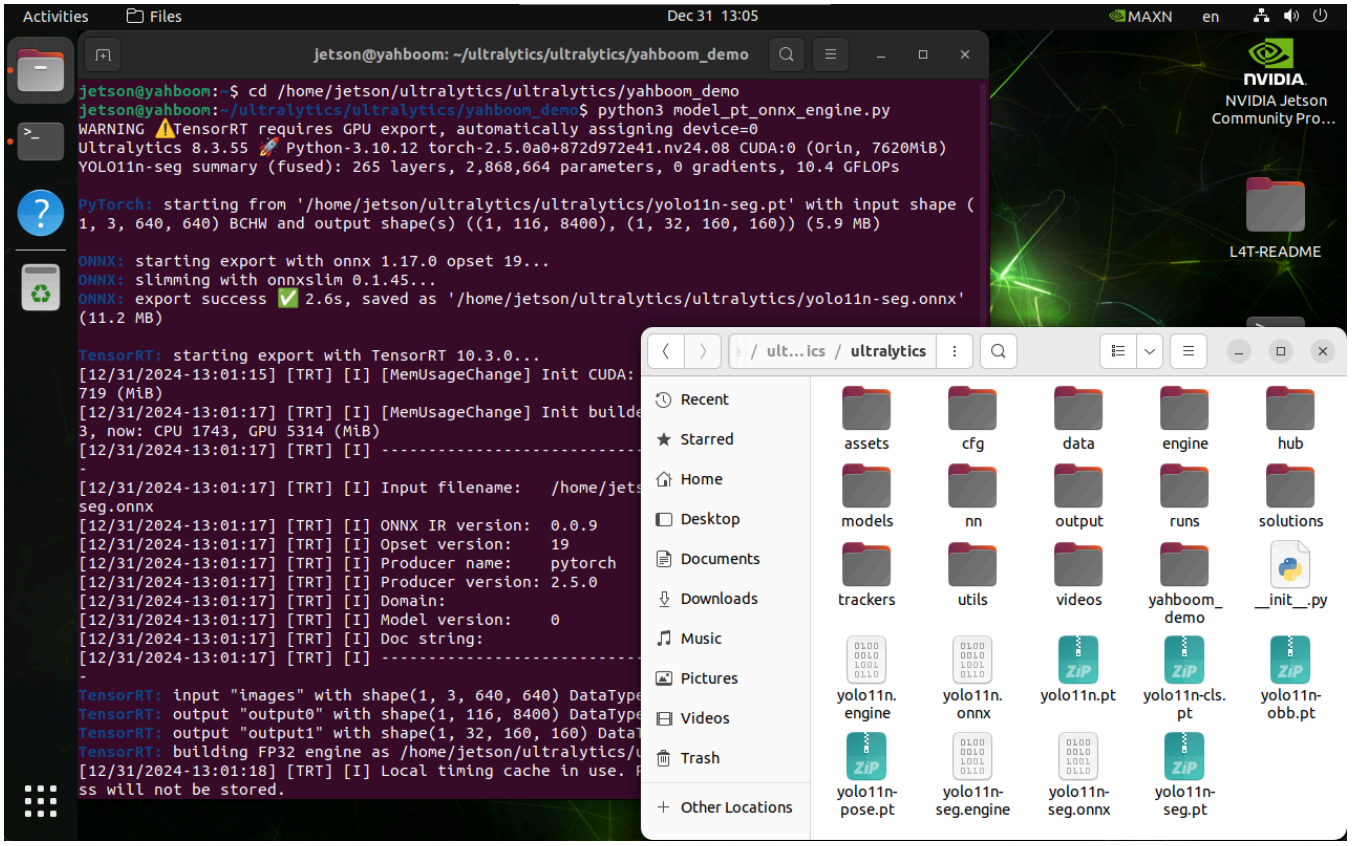
```
python3 model_pt_onnx_engine.py
```

```
from ultralytics import YOLO

# Load a YOLO11n PyTorch model
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo11n.pt")
model = YOLO("/home/jetson/ultralytics/ultralytics/yolo11n-seg.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo11n-pose.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo11n-cls.pt")
# model = YOLO("/home/jetson/ultralytics/ultralytics/yolo11n-obb.pt")

# Export the model to TensorRT
model.export(format="engine")
```

Note: The converted model file is located in the location of the converted model file



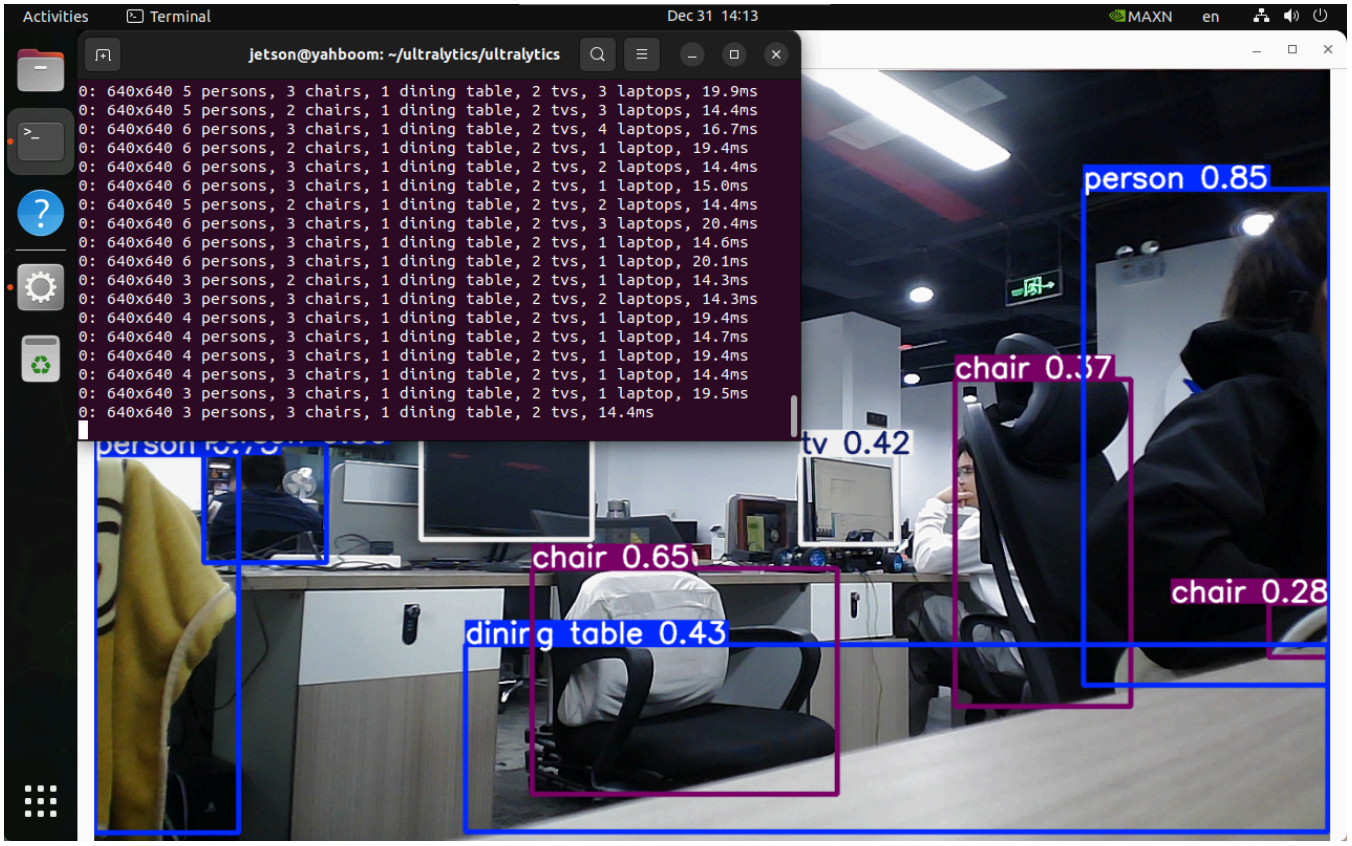
4. Model prediction

CLI usage

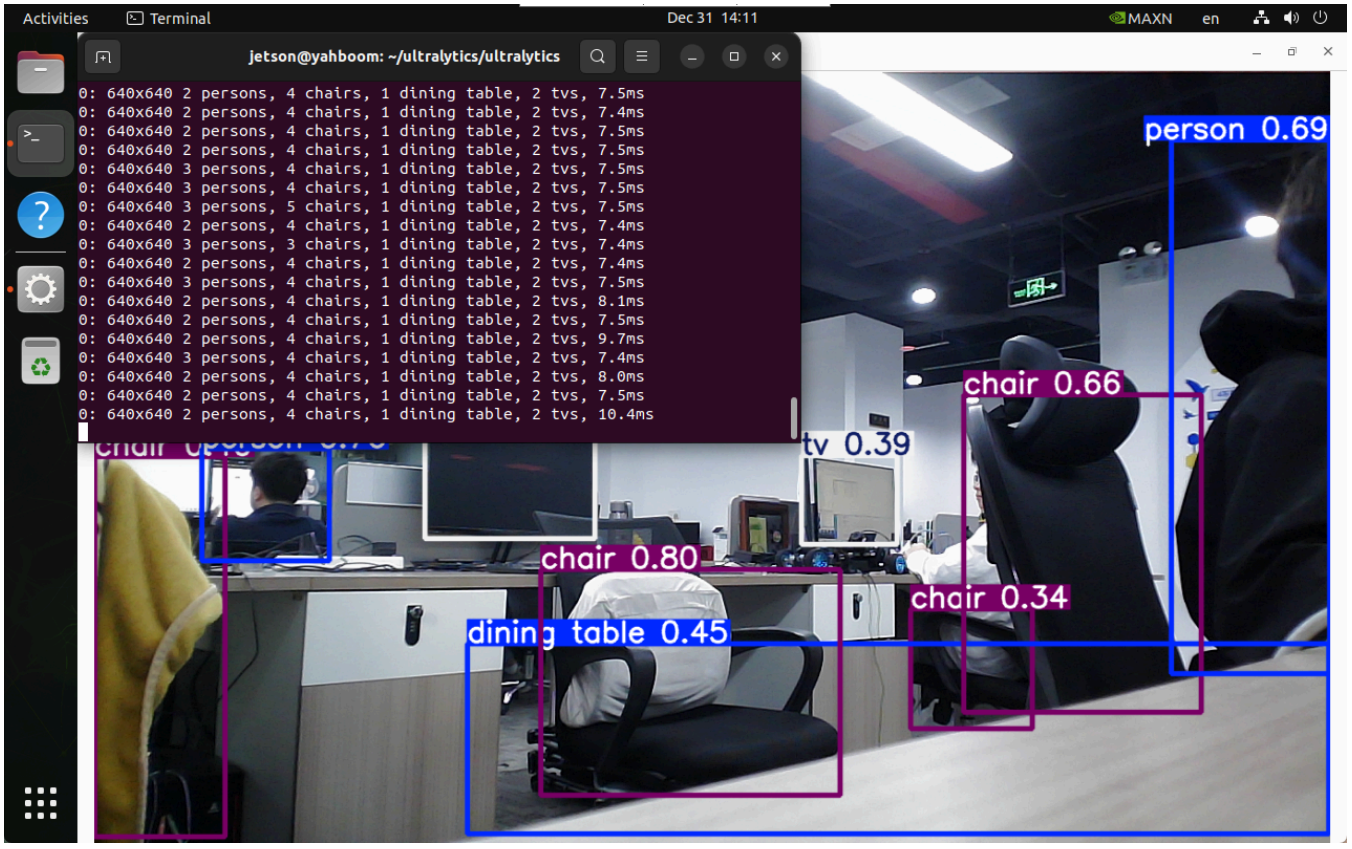
The CLI currently only supports calling the USB camera.

```
cd /home/jetson/ultralytics/ultralytics
```

```
yolo predict model=yolo11n.onnx source=0 save=False show
```

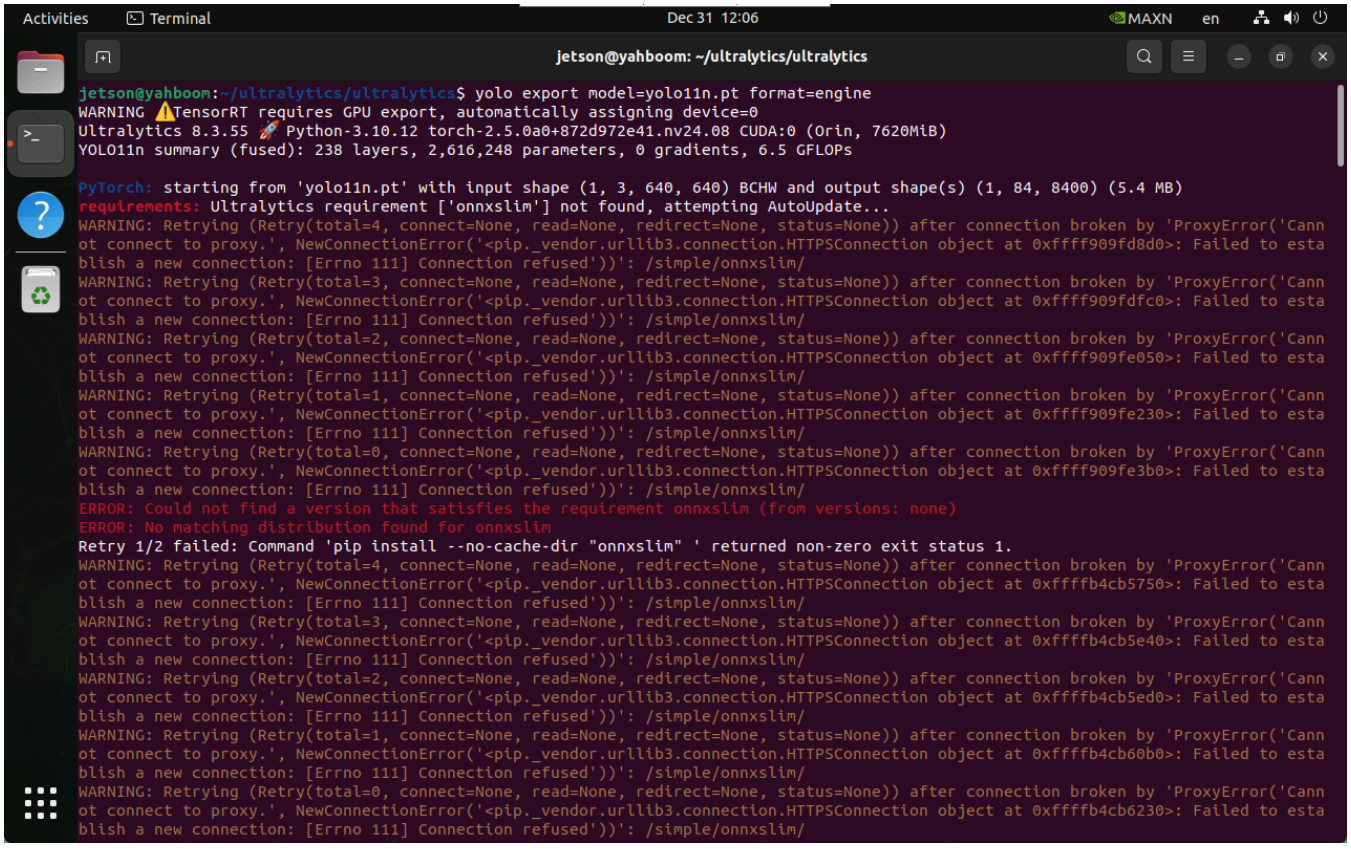



```
yolo predict model=yolo11n.engine source=0 save=False show
```



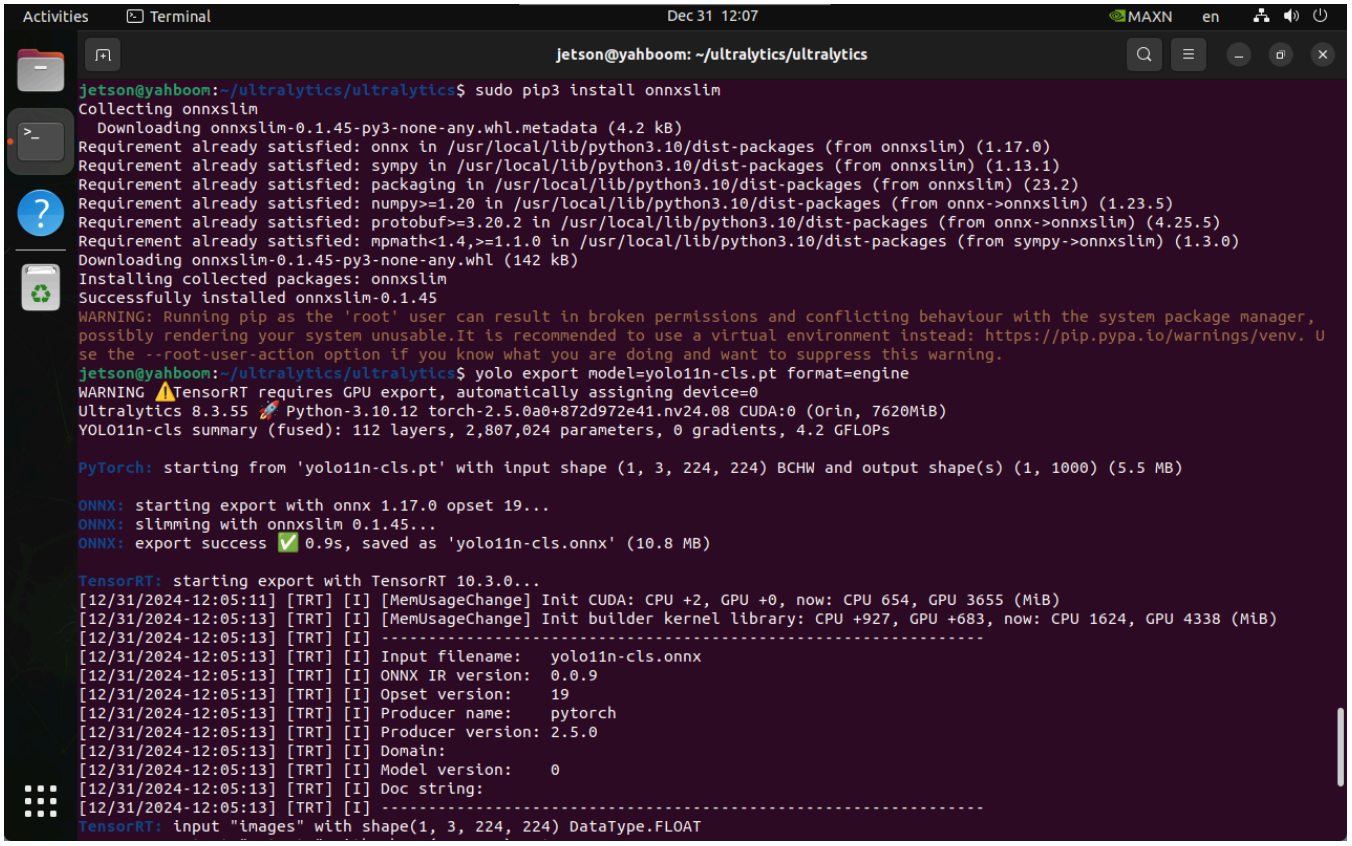
Frequently Asked Questions

ERROR: onnxslim



Solution: enter the onnxslim installation command in the terminal

```
sudo pip3 install onnxslim
```



References

https://docs.ultralytics.com/guides/nvidia-jetson/

https://docs.ultralytics.com/integrations/tensorrt/